

18

Universes of the past and future

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We will now discuss in a little more detail the struggle for existence

Charles Darwin (1809–1882)

After determining the geometry of the Robertson–Walker spaces and combining them with matter to derive the Friedmann equations, we now continue our survey of possible universes. Each of the models we discuss in this chapter makes use of the metric field described by the Robertson–Walker line element

$$ds^2 = -dt^2 + a(t)^2 \left[\frac{dr^2}{1 - kr^2} + r^2 (d\theta^2 + \sin^2 \theta d\phi^2) \right]. \quad (18.1)$$

Friedmann equation 1:

$$\left(\frac{\dot{a}}{a} \right)^2 + \frac{k}{a^2} = \frac{8\pi}{3} \rho + \frac{\Lambda}{3}.$$

Friedmann equation 2:

$$2 \frac{\ddot{a}}{a} + \left(\frac{\dot{a}}{a} \right)^2 + \frac{k}{a^2} = -8\pi p + \Lambda.$$

Recall that we also define a Hubble parameter

$$H(t) = \frac{\dot{a}(t)}{a(t)}.$$

This is a useful quantity as it tells us the rate of change of proper distance $l(t)$ via $\dot{l}(t) = H(t)l(t)$.

We choose to fill the universes with perfect fluid and vacuum energy, so can use the Friedmann equations to compute the behaviour of the expansion factor $a(t)$, which can be thought of as the effective radius of the universe. The universes can be broadly classified by the value of the curvature constant k that they take. [Recall that the two universes (numbered 0 and 1) that we have already examined had $k = 0$ and so were spatially flat.] We start this chapter by exploring some more spatially flat universes before going on to look at some $k = 1$ and $k = -1$ cases. Throughout we assume that we know the value of the expansion factor $a(t = t_0)$ at some time t_0 .

18.1 Spatially flat universes

We start by filling spatially flat ($k = 0$) universes with different sorts of matter. Recall that **Universe 0** is empty and flat. As a result, it is identical to static, Minkowski spacetime. **Universe 1** (the de Sitter model) is flat by construction, and contains only vacuum energy Λ . The result is that this latter universe expands with $a(t) = a(t_0)e^{H(t-t_0)}$, where the Hubble parameter $H \equiv \dot{a}/a = \sqrt{\Lambda/3}$ is a constant. This universe is not badly behaved at $t = 0$, where we expect a non-zero radius $a(t = 0)$ and finite mass-energy density $\rho(t = 0)$. Universe 1 does not, therefore, have any singularity at $t = 0$ and so has no Big Bang.

Now let's consider models without any vacuum energy, but where we allow cosmological fluid with a time-dependent energy density $\rho(t)$ to fill a flat universe. We then have (using the first Friedmann equation,

with $k = \Lambda = 0$) that

$$\left[\frac{\dot{a}(t)}{a(t)} \right]^2 = \frac{8\pi\rho(t)}{3} = H(t)^2, \quad (18.2)$$

where now the Hubble parameter is time dependent and is given by $H(t) = [8\pi\rho(t)/3]^{\frac{1}{2}}$. We saw in the last chapter that it is thermodynamic considerations that cause the energy density ρ to be time dependent and that $\rho(t)$ is itself a function of $a(t)$. This is to say that, as the universe evolves, the energy density ρ changes as the expansion factor $a(t)$ changes. Let's use these features to create two new, spatially flat universes. We shall fill them, not with vacuum energy density, but with either radiation (**Universe 2**) or dust (**Universe 3**).

Universe 2 is a **flat, radiation-filled** model. That is, we fill the universe with radiation, which has an energy density

$$\rho_r(t) = \left[\frac{a(t_0)}{a(t)} \right]^4 \rho_r^{(0)}, \quad (18.3)$$

where measurements are made relative to the state $\rho_r^{(0)}$ that occurs at time t_0 . The Friedmann equation becomes

$$\left[\frac{\dot{a}(t)}{a(t)} \right]^2 = \frac{8\pi\rho_r^{(0)}}{3} \left[\frac{a(t_0)}{a(t)} \right]^4. \quad (18.4)$$

For ease of writing, let's also write $a(t_0) = a_0$, and then we have

$$\dot{a}(t) = \frac{H_0 a_0^2}{a(t)}, \quad (18.5)$$

where $H_0 = [8\pi\rho_r^{(0)}/3]^{\frac{1}{2}}$. This equation is solved by an expansion factor

$$a(t) = a_0 \sqrt{2H_0 t}. \quad (18.6)$$

This spatially flat, expanding universe grows without bound as illustrated in Fig. 18.1. Notice also that at $t = 0$ we have $a(t = 0) = 0$. That is, the universe begins at zero spatial size, implying an initially infinite density of the radiation field. This is the Big Bang: an initial singularity in the model of the universe. We confirm that the density is indeed infinite in the next example.

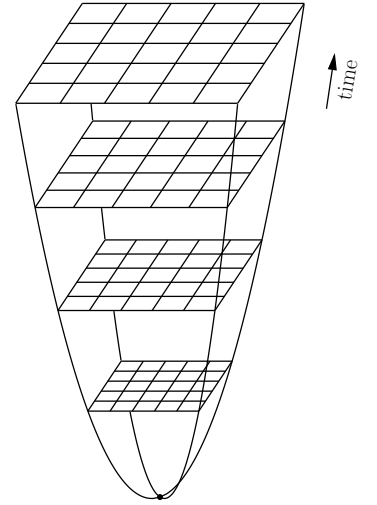


Fig. 18.1 The evolution of **Universes 2** and **3**, which have $k = 0$ and $\Lambda = 0$. These spatially flat universes expand from an initial Big-Bang singularity without bound. The flat planes represent flat three-dimensional space.

Example 18.1

With the solution in eqn 18.5, we can evaluate the Hubble parameter $H(t) = \frac{\dot{a}(t)}{a(t)}$. Plugging in, we find

$$H(t) = \frac{\frac{1}{2}a_0(2H_0/t)^{1/2}}{a_0(2H_0t)^{1/2}} = \frac{1}{2t}. \quad (18.7)$$

So we have that $H_0 = 1/2t_0$, and that the Hubble parameter (i) is badly behaved at $t = 0$, and (ii) decreases as the universe expands. This is also the case for $\rho(t)$ [since $H(t) \propto \rho(t)^{\frac{1}{2}}$], and so we deduce that, for Universe 2, the energy density starts infinite and decreases with time to zero.

Universe 3 is also known as the **Einstein–de Sitter model**. This spatially flat ($k = 0$) universe is filled only with dust, with energy density

$$\rho_d(t) = \left[\frac{a_0}{a(t)} \right]^3 \rho_d^{(0)}. \quad (18.8)$$

With this equation of state, we have a Friedmann equation

$$\dot{a} = \frac{H_0 a_0^{\frac{3}{2}}}{a^{\frac{1}{2}}}, \quad (18.9)$$

implying that

$$a(t) = a_0 \left[\frac{3}{2} H_0 t \right]^{\frac{2}{3}}. \quad (18.10)$$

From this we find that $H(t) \equiv \dot{a}/a = 2/(3t)$. Again, the universe starts from a Big-Bang singularity and grows without bound, evolving as shown in Fig. 18.1.

We have now examined three spatially flat models filled with different manifestations of energy. In each case, the universe is found to expand without limits. We summarize our findings in the table below:

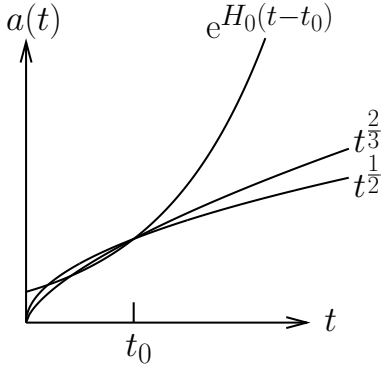


Fig. 18.2 The radius function $a(t)$ for the de Sitter (Universe 1), Radiation (Universe 2) and Einstein–de Sitter (Universe 3) models.

¹For the case of vacuum energy and dust, the more general case of this equation for $k = \pm 1$ and 0 is

$$\dot{a}^2 + k = \frac{C}{a} + \frac{\Lambda}{3} a^2, \quad (18.11)$$

which turns out to be a useful expression.

	de Sitter	Radiation	Einstein–de Sitter
Composition	Energy	Radiation	Dust
$a(t)$	$a_0 e^{H_0(t-t_0)}$	$a_0 (2H_0 t)^{\frac{1}{2}}$	$a_0 (3H_0 t/2)^{\frac{2}{3}}$
$H(t)$	const.	$1/(2t)$	$2/(3t)$
Big Bang	No	Yes	Yes

The evolution of $a(t)$ for each model is compared in Fig. 18.2.

Let's return to Universe 3 [the ($k = 0$) Einstein–de Sitter model], which contains only dust, and add in some vacuum energy Λ . The first Friedmann equation for this flat, dust-and-vacuum-energy-universe (**Universe 4**) now takes the form¹

$$\dot{a}^2 = \frac{C}{a} + \frac{\Lambda}{3} a^2, \quad (18.12)$$

where $C = \frac{8\pi}{3} a_0^3 \rho_d^{(0)}$, and we assume that the vacuum energy $\Lambda > 0$. In the initial stages of the expansion, where the radius $a(t)$ of the universe is small, we expect the dust term to be large compared to the vacuum energy term and so the universe has the behaviour described above for Universe 3, with $a \propto t^{2/3}$ (implying a Big Bang). However, with $\Lambda > 0$, at some point when $a(t)$ is large enough, the vacuum energy must dominate and we have the behaviour of the de Sitter universe (Universe 1) that $a \propto e^{Ht}$. Actually, the full behaviour of Universe 4 can be calculated exactly, with the result that

$$a(t) = \left(\frac{3C}{2\Lambda} \right)^{\frac{1}{3}} \left[\cosh \sqrt{3\Lambda} t - 1 \right]^{\frac{1}{3}}. \quad (18.13)$$

The general behaviour of this universe is therefore that it starts from a singularity and expands without bound.

Finally, we can consider the intriguing possibility (**Universe 5**) that the universe is filled with vacuum energy with a negative cosmological constant (i.e. $\Lambda < 0$). Recall that positive Λ vacuum energy exerts a negative pressure, so having $\Lambda < 0$ should be expected to provide something closer to positive-pressure mass energy, which causes gravitational attraction, and hence arrests the expansion of the universe if it is large enough. This does indeed happen: the right-hand side of the Friedmann equation vanishes when $a(t)^3 = 3C/(-\Lambda)$ and the expansion must stop. The full behaviour of the model is given by the exact solution

$$a(t) = \left[\frac{3C}{2(-\Lambda)} \right]^{\frac{1}{3}} \left[1 - \cos \sqrt{3(-\Lambda)t} \right]^{\frac{1}{3}}. \quad (18.14)$$

This solution tells us that the universe initially expands from a Big-Bang singularity, reaches a maximum radius, and then contracts, ending in the possibility of a final **Big-Crunch singularity** as shown in Fig. 18.3.

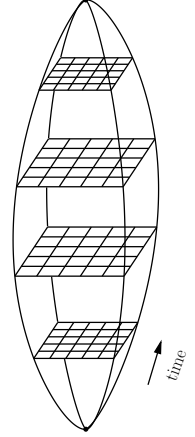


Fig. 18.3 The evolution of **Universe 5** which has $k = 0$ and negative vacuum energy.

18.2 Curved universes with $\Lambda = 0$

So far, all of the universes we have considered are spatially flat, with $k = 0$. We now turn to some spatially curved cases, and we'll start with examining the behaviour of dust-filled universes with $\Lambda = 0$. Equation 18.11, then gives us a Friedmann equation

$$\dot{a}^2 + k = \frac{C}{a}. \quad (18.15)$$

We already have the Einstein–de Sitter model of Universe 3 for the flat ($k = 0$) case, where the Universe was found to expand without bound from a singularity as shown in Fig. 18.1. What about the cases $k = \pm 1$? First, consider the case that the space is spherical ($k = 1$). This is **Universe 6** ($k = 1$, $\Lambda = 0$), examined in the next example.

Example 18.2

Start with $\dot{a}^2 = C/a - 1$ and make a change of variable $u^2 = a/C$ and the equation of motion becomes

$$\dot{u}^2 = \frac{1}{4C^2 u^2} \left(\frac{1}{u^2} - 1 \right). \quad (18.16)$$

We shall integrate this equation, employing so-called **Big-Bang initial conditions** that $a = 0$ when $t = 0$, and hence $u(0) = 0$, and we find

$$2 \int_0^{u(t)} \frac{u^2 du}{(1 - u^2)^{\frac{1}{2}}} = \frac{t}{C}. \quad (18.17)$$

The left-hand side can be integrated with the result

$$2 \int_0^{u(t)} \frac{u^2 du}{(1 - u^2)^{\frac{1}{2}}} = \sin^{-1} u(t) - u(t) [1 - u^2(t)]^{\frac{1}{2}}. \quad (18.18)$$

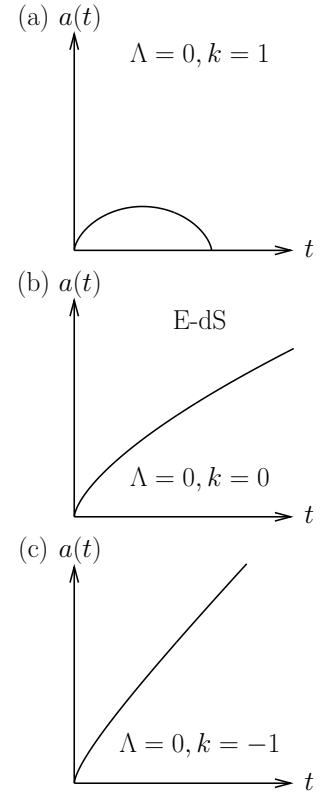


Fig. 18.4 The radius function $a(t)$ for the $\Lambda = 0$, dust-filled universes. (a) **Universe 6** with $k = 1$; (b) **Universe 3** (the Einstein–de Sitter model) with $k = 0$; (c) **Universe 7** with $k = -1$.

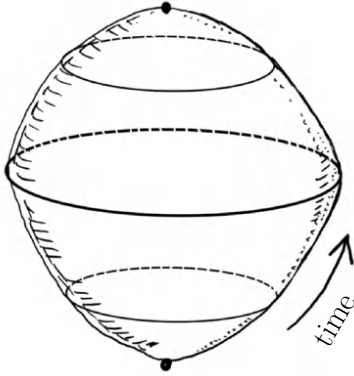


Fig. 18.5 The evolution of **Universe 6**, which has $k = 1$. The circles represent 3-spheres.

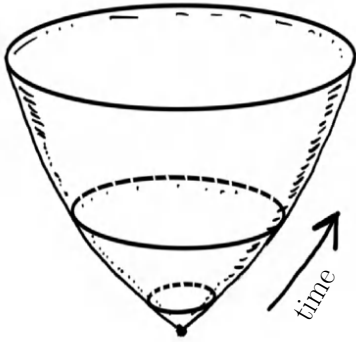


Fig. 18.6 The evolution of **Universe 7**, which has $k = -1$ and $\Lambda = 0$. The surfaces represent three-dimensional hyperbolic space.

We conclude that

$$t = C \left[\sin^{-1} \left(\frac{a}{C} \right)^{\frac{1}{2}} - \left(\frac{a}{C} \right)^{\frac{1}{2}} \left(1 - \frac{a}{C} \right)^{\frac{1}{2}} \right]. \quad (18.19)$$

The behaviour of $a(t)$ is shown in Fig. 18.4(a), where we see it expand from zero, reach a maximum and finally collapse to a zero.

The result of the last example is that Universe 6 starts from a Big Bang, expands and then collapses into a big crunch as illustrated in Fig. 18.5, where each of the spatial hypersurfaces represents a 3-sphere.

For **Universe 7**, we carry out an analogous computation for a spatially hyperbolic case ($k = -1$). We find that

$$t = C \left[\left(\frac{a}{C} \right)^{\frac{1}{2}} \left(1 + \frac{a}{C} \right)^{\frac{1}{2}} - \sinh^{-1} \left(\frac{a}{C} \right)^{\frac{1}{2}} \right]. \quad (18.20)$$

This equation, which is graphed in Fig 18.4(c), tells us that Universe 7 grows without bound. [It can be compared with the slower expansion of the Einstein–de Sitter universe shown in Fig. 18.4(b).] Universe 7 is also represented in Fig. 18.6, where each of the spatial hypersurfaces represents a hyperbolic space.

To summarize, the $\Lambda = 0$, dust-filled universes we have examined all have an initial expansion from a Big Bang. The flat ($k = 0$) case (Universe 3) and the hyperbolic $k = -1$ (Universe 7) case both grow without bound. The spherical ($k = 1$) case (Universe 6) turns over and re-collapses into a Big-Crunch singularity. The expansion factors for these universes are summarized in Fig. 18.4.

18.3 Einstein, Lemaître and Eddington

The universes so far have all had the potentially disturbing feature of being time dependent. The static **Universe 8** is constructed such that it does not have this apparent problem. This model is known as the **Einstein Universe**. It contains dust and positive vacuum energy Λ_E . We seek a *static* solution and see what happens.

Example 18.3

We start with the first Friedmann equation, written in the form for dust $\rho_d(t)$ and an amount of positive energy density Λ_E

$$\dot{a}^2 + k = \left(\frac{8\pi\rho_d}{3} + \frac{\Lambda_E}{3} \right) a^2. \quad (18.21)$$

Static implies that $\dot{a}(t) = 0$, $a(t) = a_0$ and $\rho(t) = \rho_d^{(0)}$ and so the first Friedmann equation becomes

$$k = \left(\frac{8\pi}{3} \rho_d^{(0)} + \frac{\Lambda_E}{3} \right) a_0^2. \quad (18.22)$$

It's convenient to also make use of the second Friedmann equation here with $\ddot{a} = \dot{a} = 0$, which then reads

$$\frac{k}{a_0^2} = \Lambda_E. \quad (18.23)$$

From the positivity of Λ_E we conclude that for a static universe we need to select a spatially spherical ($k = 1$) universe since this is the only way to have a positive k . The second Friedmann equation then tells us that $a_0 = \Lambda_E^{-\frac{1}{2}}$ and substituting this into eqn 18.22 we obtain $\rho_d^{(0)} = \Lambda_E/4\pi$. In words: a universe with dust density $\rho_d^{(0)}$ is kept static by the negative pressure of vacuum energy, provided that $\rho_d^{(0)} = \Lambda_E/4\pi$. It follows that the radius of the universe is given in terms of the amount of dust matter by $a_0 = (4\pi\rho_d^{(0)})^{-\frac{1}{2}}$.

The static Einstein Universe (Universe 8) is shown in Fig. 18.7 where, because $k = 1$, each spatial hypersurface represents a 3-sphere. The Einstein Universe is static, but this doesn't mean it is stable. In fact, the next example shows that it cannot withstand a small perturbation in the density of the dust that fills it.

Example 18.4

We examine the stability of this universe² by introducing a perturbation in both $a(t)$ and ρ_d . Subtracting the first Friedmann equation from the second we obtain

$$2\frac{\ddot{a}}{a} = -\frac{8\pi}{3}\rho_d + \frac{2}{3}\Lambda_E. \quad (18.24)$$

We perturb around the equilibrium values by setting $a = \Lambda_E^{-\frac{1}{2}} + \delta a$ and $\rho_d = \frac{\Lambda_E}{4\pi} + \delta\rho_d$, giving, to order δa , the equation of motion

$$2\Lambda_E^{\frac{1}{2}}(\ddot{\delta a}) = -\frac{8\pi}{3}\delta\rho_d. \quad (18.25)$$

If the total amount of dust is constant, then we also have the thermodynamic result that

$$\rho_d a^3 = \text{const.}, \quad (18.26)$$

and so we can find an equation for $\delta\rho_d$ by differentiating:

$$\frac{\delta\rho_d}{\rho_d} = -3\frac{\delta a}{a}. \quad (18.27)$$

Substituting into eqn 18.25, we conclude that

$$(\ddot{\delta a}) = \Lambda_E(\delta a), \quad (18.28)$$

whose solution is $\delta a \propto e^{\sqrt{\Lambda_E}t}$. The result, therefore, is that the perturbation grows exponentially in time, causing the universe to evolve. The static universe is not in a stable equilibrium.

We conclude that the Einstein Universe is only just static, since it contains a critical amount of energy density $\Lambda_E = 4\pi\rho_d$. The vacuum energy tries to make the universe expand; the matter tends to cause the universe to contract and they end up in unstable equilibrium.

In **Universe 9**, known as the **Lemaître solution**,³ we have $k = 1$ and a universe full of dust and vacuum energy, but this time with $\Lambda > \Lambda_E$. Since the vacuum energy tends to make universes expand, we should expect that this universe expands without bound. However, there is a surprise: the model has a **coasting period** where it stays at roughly the same size. This coasting period can last for arbitrarily long time if the energy density is set appropriately. This is represented in Fig. 18.8, where each spatial hypersurface represents a 3-sphere.

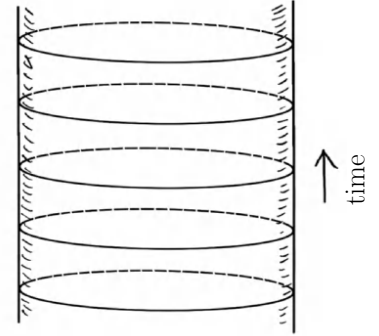


Fig. 18.7 The evolution of **Universe 8**, the Einstein Universe, which has $k = 1$ and vacuum energy $\Lambda = \Lambda_E$. Each circle represents a 3-sphere.

²Here we follow the approach described in the problem book by Lightman *et al.*

³George Lemaître (1894–1966)

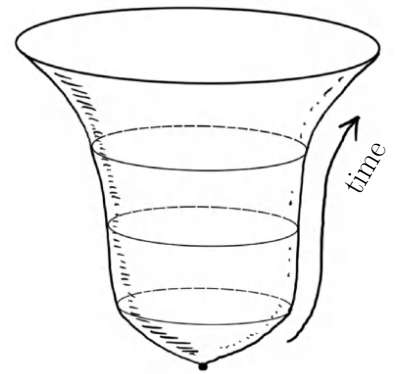


Fig. 18.8 The evolution of **Universe 9** which has $k = 1$ and vacuum energy $\Lambda > \Lambda_E$.

⁴The algebraic steps are examined in Exercise 18.4.

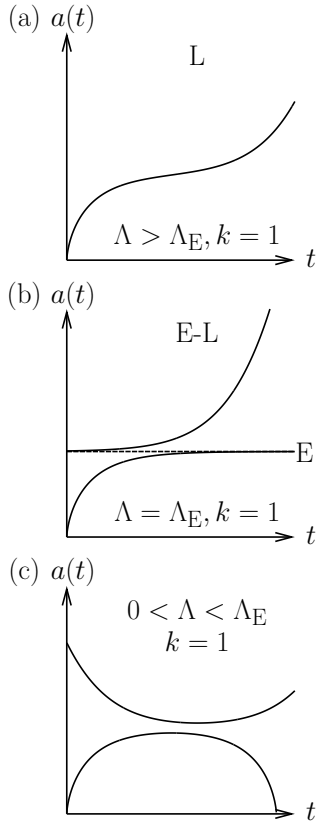


Fig. 18.9 The radius function $a(t)$ for the some $k = 1$ universes. (a) **Universe 9** (the Lemaître model) with $\Lambda > \Lambda_E$ showing its famous coasting period; (b) **Universe 8** (the Einstein static universe with Λ_E) and **Universe 10** (The Eddington–Lemaître model), in its two forms, distinguished by different initial conditions; (c) **Universe 11**, with $0 < \Lambda < \Lambda_E$ for two distinct initial conditions.

Example 18.5

Consider Universe 9 for the case where $\Lambda = (1 + \varepsilon)^2 \Lambda_E$, where ε is assumed small. Since, for the Einstein Universe we have $a_0 = (4\pi\rho_d)^{-\frac{1}{2}}$ and $a_0 = \Lambda_E^{-\frac{1}{2}}$, we can write for the dust that

$$\frac{8\pi\rho(t)}{3} = \frac{2}{3\Lambda_E^{\frac{1}{2}}a(t)^3}. \quad (18.29)$$

Substituting $\Lambda/(1 + \varepsilon)^2$ for Λ_E and setting $k = 1$, we write the first Friedmann equation in the form

$$\dot{a}^2 + 1 = V_{\text{eff}}(a), \quad (18.30)$$

where we've defined an effective potential

$$V_{\text{eff}}(a) = \frac{2(1 + \varepsilon)}{3\Lambda_E^{\frac{1}{2}}a} + \frac{\Lambda}{3}a^2. \quad (18.31)$$

From the last example we see that when a is small, the radius of Universe 9 varies as $\dot{a} \propto 1/a^{\frac{1}{2}}$. We examined this behaviour in the Einstein–de Sitter model (Universe 3) and found that it implies that the system expands from an initial singularity as $a \propto t^{\frac{2}{3}}$. However, as a becomes larger the rate of expansion slows, reaching a minimum when $\dot{a}$ is a minimum. This is the case when⁴

$$a = a_{\min} = \frac{(1 + \varepsilon)^{\frac{1}{3}}}{\Lambda_E^{\frac{1}{2}}}. \quad (18.32)$$

The universe then **coasts** for a period, before speeding up again.

Example 18.6

Expanding the effective potential $V(a)$ about the minimum, we have

$$\begin{aligned} \dot{a}^2 &\approx -1 + V(a_{\min}) + \frac{1}{2}(a - a_{\min})^2 \left. \frac{\partial^2 V}{\partial a^2} \right|_{a=a_{\min}} \\ &= -1 + (1 + \varepsilon)^{\frac{2}{3}} + \left[\Lambda_E^{\frac{1}{2}}a - (1 + \varepsilon)^{\frac{1}{3}} \right]^2. \end{aligned} \quad (18.33)$$

This equation can be integrated. The result (a standard integral that can be looked up) is

$$a(t) = \frac{(1 + \varepsilon)^{\frac{1}{3}}}{\Lambda_E^{\frac{1}{2}}} \left\{ 1 + \left[1 - (1 + \varepsilon)^{-\frac{2}{3}} \right]^{\frac{1}{2}} \sinh \left[\Lambda_E^{\frac{1}{2}}(t - t_m) \right] \right\}, \quad (18.34)$$

where t_m is the time at which $\dot{a}$ reaches its minimum. In the limit of small ε , we have

$$a \approx a_{\min} \left[1 + \frac{\varepsilon}{3} \sinh \Lambda_E^{\frac{1}{2}}(t - t_m) \right]. \quad (18.35)$$

As a result, a is approximately $a_{\min}$ until

$$\varepsilon \sinh \Lambda_E^{\frac{1}{2}}(t - t_m) \approx 1, \quad (18.36)$$

which is to say, for a time

$$(t - t_m) \approx \frac{\ln \left(\frac{1}{\varepsilon} \right)}{\Lambda_E^{\frac{1}{2}}}. \quad (18.37)$$

This coasting period can therefore be made arbitrarily long with a sufficiently small ε . Eventually, the expansion starts again with $(\dot{a}/a)^2 \rightarrow \Lambda/3$ as we had for the de Sitter model (Universe 1).

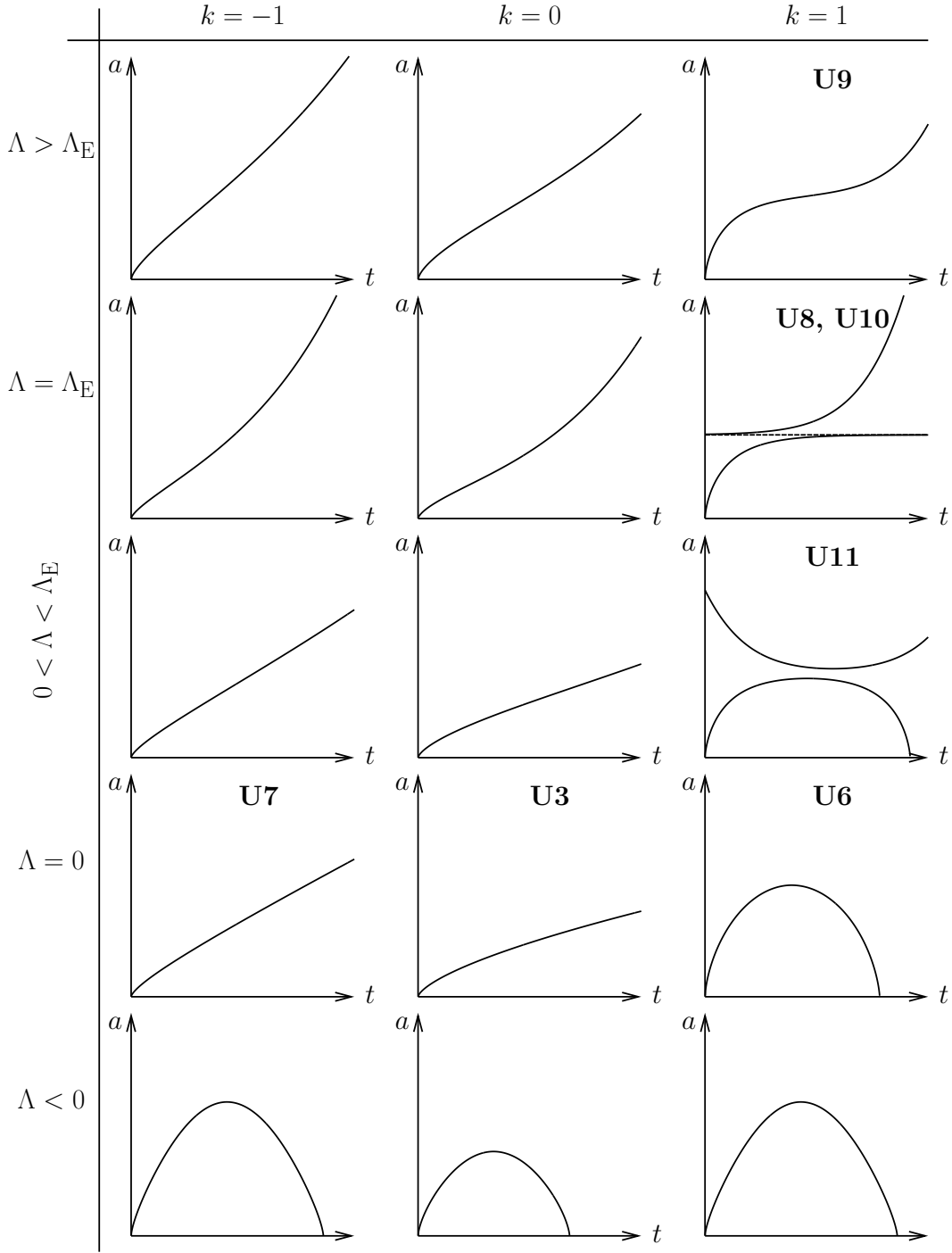


Fig. 18.10 The origins and fates of some Robertson-Walker spacetimes. These were computed for dust and vacuum energy universes using eqn 18.11.

⁵Arthur Eddington (1882–1944) was an early champion of general relativity, writing several articles describing it for an English-speaking audience, who were cut off from many developments in Germany owing to World War I.

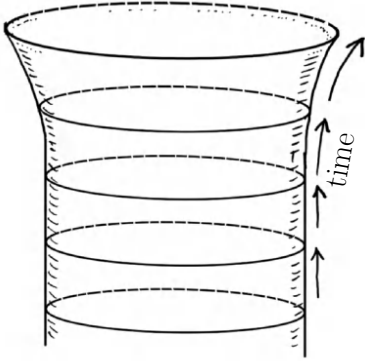


Fig. 18.11 Universe 10, the Eddington-Lemaître model, starts similarly to the Einstein static universe and begins to expand after a certain time.

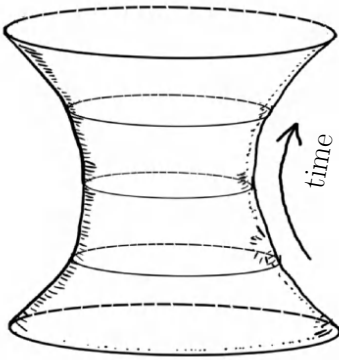


Fig. 18.12 Universe 11, showing an initial contraction, followed by a re-expansion.

The behaviour of the radius function in Universe 9 is shown in Fig. 18.9(a).

As a result of studying the Lemaître model, Arthur Eddington⁵ proposed a model without a Big-Bang singularity. Instead the model starts by coasting at a value of a close to the static Einstein value until, at some point, it starts expanding. This is **Universe 10: the Eddington-Lemaître model** shown in Fig. 18.9(b) and Fig. 18.11. Actually, there is another variant of this model: we could start from a Big-Bang singularity and asymptotically approach the Einstein Universe as shown in Fig. 18.9(b).

Finally, what if we have $0 < \Lambda < \Lambda_E$? This is the content of **Universe 11**. In this case, there are two possibilities, depending on the initial conditions. If we start from an initial Big-Bang singularity then there will not be enough vacuum energy to prevent the matter collapsing back in itself and we have a Big Crunch singularity. Perhaps more interestingly, if we start at a very large value of a , then the universe contracts, reaches a minimum radius, and then expands again. It has $a \propto e^{H(-t)}$ as $t \rightarrow -\infty$ and $a \propto e^{Ht}$ as $t \rightarrow \infty$. This is shown in Fig. 18.12 with a radius function as shown in Fig. 18.9(c).

We can summarize some of our progress by examining the evolution of several universes in Fig. 18.10.

18.4 A brief history of model universes

We have presented several model universes generated by varying the content and geometry of the models described by the Friedmann equations. This was not the order in which they were discussed historically. The discovery that our Universe is expanding was established through observations made by a number of people in the decades following 1910. Perhaps most famous of the astronomers in this period is Edwin Hubble, who in 1929 established that the speed of recession of a galaxy v_d is proportional to its distance from us. This is Hubble's law that $v_d = H \times \text{distance}$.

Somewhat earlier, back in 1917, Einstein had considered filling a spatially spherical universe with uniform matter and realized that, owing to the gravitational attraction of the matter, the resulting model could not be static without the incorporation into the theory of an extra energy contribution Λ , that he called the cosmological term. In the same year, de Sitter worked out that, for an isotropic model, the cosmological-constant term could provide a repulsion that exactly cancelled the gravitational attraction. This gives us the Einstein model (Universe 8). The model ceased to be considered seriously after it became established that the physical Universe was expanding. Eddington in 1930 showed that this universe was unstable to perturbation and, in 1931, Einstein recommended the cosmological term be dropped from general relativity.

In 1917, de Sitter had also observed that he could model an expanding universe if he included only the Λ term. Alexander Friedmann derived his results for evolving isotropic and homogenous universes in 1922, which also suggested the Hubble-law expansion $v_d = H \times \text{distance}$. (In 1923, Hermann Weyl pointed out that test particles in the de Sitter model would separate at a rate consistent with this law.) In 1927, the Catholic priest and physicist George Lemaître independently derived Friedmann's homogeneous, isotropic models and proposed the Eddington–Lemaître model (Universe 10). This suggestion of a model that started at the Einstein solution and expanded was promoted by Eddington. However, Lemaître would in 1931 suggest that the universe could have expanded from what he termed a ‘Primeval Atom’ (his Universe 9), an idea later described rather dismissively by Fred Hoyle⁶ as the ‘Big Bang theory’. Robertson and Walker would independently show in 1935 that the line element Lemaître was using represents the most general geometry compatible with homogeneity and isotropy.

There was a significant problem with models involving the expansion of the universe from an initial point in spacetime. The age of rocks (and hence the Earth) had been estimated from the radioactive properties of uranium to be at least 3×10^9 years, but this didn't seem to allow long enough for the universe to evolve into its current state.

⁶Fred Hoyle (1915–2001) rejected the Big Bang theory and supported an alternative steady-state picture, in which there is no cosmological constant, but where a creation field (or C-field) is responsible for matter generation as the Universe expands, allowing the density to remain constant in time. The C-field has negative energy density (allowing for conservation of energy as new matter is created) and so exerts a negative pressure. The inability of the steady-state model to account for the cosmic microwave background, discovered in 1964, is one of the reasons why it was abandoned by cosmologists. The term ‘Big Bang’ was coined in a talk given by Hoyle on BBC radio in 1949. Hoyle denied that he intended to be insulting; rather that he was emphasizing the difference between the models for a general audience.

Example 18.7

Let's start by assuming that the acceleration of our Universe has always been negative (i.e. the rate of expansion has always been slowing down). If we consider the evolution of $a(t)$ then its tangent evaluated now $\dot{a}(t_{\text{now}})$ cuts the axis at an earlier time t_1 where

$$a(t_{\text{now}}) = \dot{a}(t_{\text{now}})(t_{\text{now}} - t_1). \quad (18.38)$$

This is shown in Fig. 18.13. Rearranging, we find

$$t_{\text{now}} - t_1 = \frac{a(t_{\text{now}})}{\dot{a}(t_{\text{now}})} = H(t_{\text{now}})^{-1}. \quad (18.39)$$

Before the 1950s the quantity $H(t_{\text{now}})^{-1}$ was estimated to be 1.8×10^9 years. If the universe does start at, or around, $a(t = 0) = 0$, then its age must be less than $H(t_{\text{now}})^{-1}$.

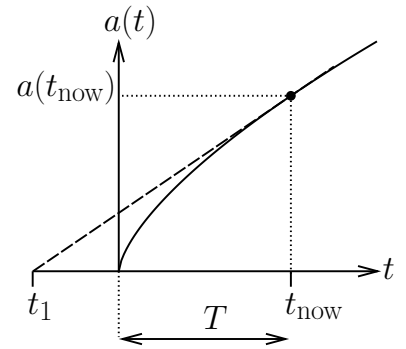


Fig. 18.13 Estimation of the age T of the Universe using $a(t)$. Assuming the Universe has been monotonically decelerating, we must have $T < t_{\text{now}} - t_1$.

If the ages of the rocks are as claimed on the basis of the nuclear physics, then there doesn't seem to have been enough time for the Universe to evolve before the creation of rocks. This is known as the **time-scale problem**. One solution was the introduction into the Lemaître model of large positive values of Λ that cause a large acceleration. It is also solved by the Eddington–Lemaître model, since this starts from the Einstein solution before expansion. However, in that model the assumption that the stars condense at the moment that the expansion starts is problematic, since it is not clear how this can happen within a static and homogeneous model.

By the 1930s, nuclear physics was also offering explanations of the genesis of the elements (nucleogenesis) which seemed consistent with Big

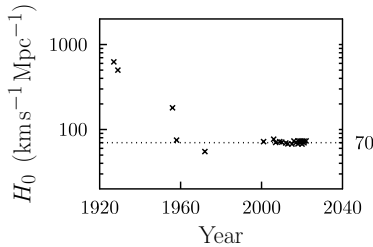


Fig. 18.14 Selected measurements of H_0 .

↪ The evolution of the very early Universe is discussed in Chapter 41. See Chapter 15 for a brief description of the anisotropy of the CMB.

⁷Recent estimates give h to be close to 0.7 (see Fig. 18.14).

↪ In Section 49.8 we will discuss our current understanding, based on experimental evidence, on the question of which type of Universe we are actually living in.

Bang models, since as the scale factor a tends to zero, the temperature should go as $T \propto a^{-1}$ allowing, in the early moments after the Big Bang, the extreme conditions needed to fuse hydrogen to make the heavier elements. This led Lemaître to promote the idea of a **hot Big Bang**.

By the 1950s new measurements of Hubble's constant H_0 (see Fig. 18.14 for a timeline) suggested an increased lifetime of the Universe of 1.3×10^{10} years, effectively resolving the timescale problem without the need for a large cosmological constant, so that Big Bang models with $\Lambda = 0$ could now be seriously considered. This had an aesthetic appeal, owing to the slightly ad-hoc nature of the introduction of the cosmological constant. The $\Lambda = 0$ models became known as the **standard models** of cosmology. These are, for $k = 0$, Universe 3 (Einstein–de Sitter), for $k = 1$, Universe 6 and for $k = -1$, Universe 7. They are all Big Bang models.

Compelling evidence for a Big-Bang cosmology came from the measurement of the cosmic microwave background (CMB), which is the faint electromagnetic radiation remaining from recombination events that occurred during the early stages of the Universe. The existence of the CMB had originally been predicted in 1948 by Ralph Alpher and Robert Herman, following work by George Gamow. Alpher and Herman predicted that high-energy radiation from the very early Universe should have been shifted into the microwave region of the electromagnetic spectrum, and that this should result in an effective black-body temperature of space of $T = 5$ K. The CMB was accidentally discovered by Arno Penzias and Robert Wilson in 1964, and has since been subject to very precise measurements, that determine its temperatures to be 2.73 K, with faint anisotropies.

Which of the three standard models our Universe most resembles is determined by how the mass-energy content of the Universe compares with the critical density (discussed in the last chapter) of $\rho_c = 3H_0^2/8\pi$. Expressing the Hubble constant⁷ as $H_0 = h \times 100 \text{ km s}^{-1} \text{ Mpc}^{-1}$, the critical density is $\rho_c = 1.88 \times 10^{-26} h^2 \text{ kg m}^{-3}$. Observations suggests that the density of observable matter in the Universe is actually only a small fraction of this, at around $10^{-28} \text{ kg m}^{-3}$. However, observations also suggest that the properties of the Universe are consistent with a flat, $k = 0$ model. This discrepancy is known as the **missing mass problem** and remains one of the key questions about our knowledge of the Universe.

So far we have been quite cavalier in accepting the presence of the Big-Bang singularity. We are now at the point where we must engage with it, and the break down of physical laws that it brings. This is the goal of the next chapter.

Chapter summary

- The Robertson–Walker spaces filled with a perfect fluid provide a range of models, many of which start with a Big Bang.
- The Einstein Universe is static, but unstable; the Einstein–de Sitter model expands without bound and the Lemaître model has a coasting period.
- The standard models of cosmology have zero cosmological constant Λ , and all begin with a Big-Bang singularity.

Exercises

- (18.1) Let's examine a Robertson–Walker universe with no matter or vacuum energy in it, by letting $\rho = p = \Lambda = 0$.

(a) Set $k = -1$ and show that the resulting metric line element can be written as

$$ds^2 = -dt^2 + t^2 [d\chi^2 + \sinh^2 \chi (d\theta^2 + \sin^2 \theta d\phi^2)]. \quad (18.40)$$

(b) Using the transformations

$$r = t \sinh \chi, \quad T = t \cosh \chi \quad (18.41)$$

show that this is a familiar spacetime in disguise.

- (18.2) Verify by substitution that (a) eqn 18.13 and (b) eqn 18.14 are solutions to the Friedmann equations.

- (18.3) Verify eqns 18.25 and 18.28.

- (18.4) Verify eqns 18.32 and 18.33.

*There is a neat geometrical interpretation of the **de Sitter universes**, which are those that are driven only by a non-zero cosmological constant Λ . (Universe 1 is the $k = 0$ version. There are also $k = \pm 1$ versions.) The construction involves embedding a hyperboloid*

$$-T^2 + W^2 + X^2 + Y^2 + Z^2 = \alpha^2, \quad (18.42)$$

with α a constant, in a $(4+1)$ -dimensional Minkowski space with line element $ds^2 = -dT^2 + dX^2 + dY^2 + dZ^2 + dW^2$. The hyperboloid is shown in Fig. 18.15, with two of the dimensions suppressed (i.e. the circular cuts should be regarded as 3-spheres in five dimensions). This seems very

abstract, by the idea of covering the hyperboloid with different coordinate system to see the features of the physics, can be demonstrated on the two-dimensional surface of a hyperboloid described by $-T^2 + X^2 + Y^2 = 1$, embedded in $(2+1)$ -dimensional Minkowski space, as demonstrated by the following exercises.

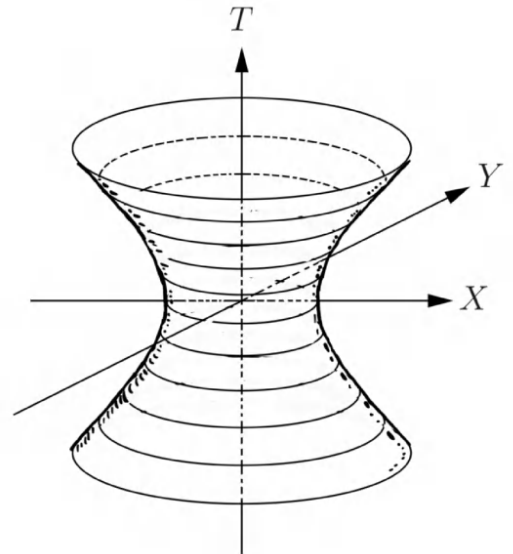


Fig. 18.15 de Sitter spacetime represented as a hyperboloid.

- (18.5) As a warm-up exercise, use the method from Appendix D to embed the hyperbola $-T^2 + X^2 = 1$ in (1+1)-dimensional Minkowski space with line element $ds^2 = -dT^2 + dX^2$.
- (18.6) Embed the hyperbolic surface $-T^2 + X^2 + Y^2 = 1$ shown in Fig. 18.15 in Minkowski space with line element $ds^2 = -dT^2 + dX^2 + dY^2$ by eliminating the variable Y , to show
- (a) the line element can be written in terms of variables v and ψ as

$$ds^2 = -\frac{dv^2}{1+v^2} + v^2 d\psi^2; \quad (18.43)$$

(b) the latter expression can be reduced to the hyperbolic line element

$$ds^2 = -d\chi^2 + \sinh^2 \chi d\psi^2, \quad (18.44)$$

where you should determine the relationship between coordinates (χ, ψ) and (T, X, Y) .

(c) Compare the latter with the other form of the line element from the chapter, which was

$$ds^2 = -d\chi^2 + \cosh^2 \chi d\theta^2. \quad (18.45)$$

At constant timelike variable χ there is a close resemblance between (i) eqn 18.44 and the line element of a spherical universe with $a(t) = \sinh t$ and (ii) eqn 18.45 and the line element of a hyperbolic Universe with $a(t) = \cosh t$.

(d) Confirm this is the case by considering the Friedmann equations.

Notice how that, by choosing coordinates that split the space and time up in different ways, we are able to use de Sitter spacetime to represent cosmological-constant-driven models with different spatial curvatures. This works because the different cross sections of the hyperboloid represented by each coordinate system possess different spatial curvatures.

- (18.7) Start again with a hyperbolic surface $-T^2 + X^2 + Y^2 = 1$ embedded in (2+1)-dimensional Minkowski space in the form that results from eliminating Y :

$$ds^2 = -dT^2 + dX^2 + \frac{(TdT - XdX)^2}{(1 + T^2 - X^2)}. \quad (18.46)$$

Now define variables

$$\begin{aligned} T &= \sinh t + \frac{x^2}{2}e^t, \\ X &= \cosh t - \frac{x^2}{2}e^t, \end{aligned} \quad (18.47)$$

implying $Y = xe^t$.

(a) Show that the line element becomes

$$ds^2 = -dt^2 + e^{2t} dx^2, \quad (18.48)$$

which is the line element for the spatially expanding de Sitter Universe (Universe 1) in (1+1) dimensions.

(b) Show that these coordinates only cover half of the hyperboloid.

(c) What kind of fixed-time slices do these coordinates make on the hyperboloid?

(d) Find a transformation that turns eqn 18.48 into the line element

$$ds^2 = \frac{1}{u^2} (-du^2 + dx^2). \quad (18.49)$$

The latter is a type of Poincaré line element for Minkowski space, demonstrating that the Poincaré line element describes a hyperbolic spacetime.

- (18.8) Using the same method as in Exercise 18.7, show that a description of (1+1)-dimensional de Sitter spacetime using coordinates

$$\begin{aligned} T &= (1-x)^{\frac{1}{2}} \sinh t, \\ X &= (1-x^2)^{\frac{1}{2}} \cosh t, \end{aligned} \quad (18.50)$$

gives the Schwarzschild-like static line element

$$ds^2 = -(1-x^2)dt^2 + \frac{dx^2}{1-x^2}. \quad (18.51)$$

How much of the hyperboloid is covered by these coordinates?

The key feature of this coordinate system is that it is static. Since a cosmological constant is time-independent, this is not really a surprise.